

Input Representation Dominates Model Choice in Machine-Learning Modulation Format Identification

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Abstract—Modulation format identification (MFI) lets a coherent optical receiver determine the incoming format without a control channel, and is a core function of optical performance monitoring in elastic optical networks. Convolutional neural networks (CNNs) applied to constellation diagrams are known to outperform classical feature-based classifiers, but the reported comparisons change two variables at once: the learning algorithm becomes deep, and the input becomes a two-dimensional image. It is therefore unclear whether the reported gains come from the model or from the representation. We disentangle the two on a common simulated benchmark of 63,000 constellation captures spanning 5 formats (QPSK, 8-QAM, 16-QAM, 32-QAM, 64-QAM) and OSNR from 5 to 25 dB, generated with a channel model that includes laser phase noise, residual frequency offset, I/Q imbalance and residual chromatic dispersion in addition to ASE noise. We evaluate five classifiers over three input representations: raw I/Q samples, a 64×64 constellation density image, and 15 higher-order statistical features. Holding the representation fixed and varying the algorithm across a multilayer perceptron, a support vector machine and a k -nearest-neighbour classifier changes accuracy by only 1.83 percentage points; holding the algorithm class fixed and changing the representation from features to the density image changes it by 5.32 points. The image-based CNN attains $98.73 \pm 0.12\%$ overall and reaches 95% accuracy 7 dB earlier in OSNR than the best feature-based model. The gap widens sharply under distribution shift: trained on an AWGN-only channel and tested on a severely impaired one, the CNN retains 89.78% while the feature-based classifier falls to 21.84%, essentially the 20% chance level. We further show that the learned model interpolates to OSNR values withheld from training, that a 512-symbol capture processed as an image outperforms a 4096-symbol capture reduced to features, and that accuracy is insensitive to model capacity across a $9.2\times$ range of parameter counts. The results indicate that, for this task, effort is better spent on the input representation than on model capacity — and that reporting only matched-channel accuracy can conceal a complete loss of function under realistic deployment conditions.

Index Terms—Modulation format identification, optical performance monitoring, convolutional neural network, constellation diagram, coherent detection, machine learning.

I. INTRODUCTION

Elastic optical networks adapt the modulation format of each lightpath to the conditions of the link, using dense formats such as 64-QAM on short, clean routes and robust formats such as QPSK on long, noisy ones. Before a coherent receiver can demodulate anything it must know which format has arrived, because the decision rules differ for each. Conven-

tionally this is negotiated through a control channel. *Modulation format identification* (MFI) removes that dependency by inferring the format from the received signal itself, enabling faster reconfiguration and more autonomous operation, and it is a standard component of optical performance monitoring (OPM) [1].

Classical MFI relies on hand-crafted statistics of the received samples, most commonly higher-order cumulants, which have the useful property that the contribution of Gaussian noise cancels at fourth order and above [2], [3]. These features are fed to a conventional classifier such as a support vector machine (SVM) or a k -nearest-neighbour (k -NN) rule. More recently, deep learning has been applied directly to the constellation diagram, treating MFI as an image classification problem, with reported gains over the classical pipeline [4]–[6].

The comparison underlying these gains, however, varies two factors at once. A CNN operating on a constellation image differs from an SVM operating on cumulants both in the model, which is deep and learned rather than shallow and kernel-based, and in the input, which is a two-dimensional density map rather than a short vector of summary statistics. In the wireless AMC literature this confound has been addressed by holding the classifier fixed and varying the representation [7]; in the optical domain, where the impairments differ substantially, we are not aware of an equivalent controlled comparison.

This paper separates them by holding one factor fixed while varying the other. We find that the representation dominates: three learning algorithms of very different type, given identical features, span 1.83 percentage points, whereas changing the input to a density image gains 5.32 points. The difference is larger still under channel mismatch, where the feature-based classifier loses essentially all discriminative ability while the CNN degrades gradually. Our contributions are:

- 1) A controlled comparison of five classifiers across three input representations on identical signals, which separates the effect of the *model* from the effect of the *representation*. In particular we add a multilayer perceptron (MLP) on the same 15 features given to the SVM and k -NN, so that “deep model” and “image input” are varied independently.

- 2) A channel model that goes beyond additive white Gaussian noise (AWGN) to include laser phase noise with block-average carrier phase recovery, residual frequency offset, receiver I/Q gain imbalance and phase skew, and residual chromatic dispersion.
- 3) Evidence that the learned model generalises to OSNR levels withheld from training, addressing the concern that such models merely memorise the appearance of each noise level.
- 4) An accuracy–complexity study, together with the shortest capture length that still supports reliable identification. Both quantities constrain a monitoring receiver with a limited compute and memory budget, and neither is commonly reported.
- 5) A cross-channel robustness study quantifying the loss incurred when a model trained on one channel model is applied to a more heavily impaired one. We find this loss to differ by an order of magnitude between the two representations.

All code, data-generation scripts and trained models are released, and every number reported here is reproducible from a fixed random seed.

II. RELATED WORK

Automatic modulation classification has a long history in wireless communications, where cumulant-based decision trees and feature classifiers were established well before deep learning [2], [3]. O’Shea *et al.* [6] showed that convolutional networks operating on raw sampled waveforms could match or exceed expert-feature approaches for radio signals.

In the optical domain, Khan *et al.* [5] applied deep neural networks to asynchronous amplitude histograms for joint OSNR monitoring and format identification, and Wang *et al.* [4] introduced the “intelligent constellation diagram analyser”, in which a CNN classifies constellation images and simultaneously estimates OSNR. A broad survey of machine learning for optical communications is given in [1].

The question of input representation has been addressed directly in the *wireless* AMC literature. Ge *et al.* [7] compare five representations — higher-order cumulants, fuzzy c -means features, grid-like constellation diagrams, cumulative distribution functions and raw I/Q — while holding the classifier fixed, and report that cumulants, raw I/Q and grid constellations perform comparably under a Gaussian channel, whereas cumulant performance degrades markedly under non-Gaussian conditions. A recent survey [9] organises the DL-AMC literature explicitly by data representation. Our study is complementary in three respects: it addresses the optical channel, in which OSNR and transceiver impairments rather than multipath dominate; it varies the algorithm as well as the representation, so that both directions of the comparison are controlled; and it reaches a different conclusion for the matched-channel case, where we find the density image to be clearly superior rather than comparable.

Within optical MFI, image-derived features have been used with conventional classifiers: Huang *et al.* [8] extract fractal-

dimension and grey-level co-occurrence features from constellation diagrams and classify them with an SVM, reporting 100% accuracy above 10 dB OSNR for seven formats. That work varies the representation but not the classifier, and evaluates under a single channel condition.

The cross-channel evaluation of Section V-F, in which a model trained under one impairment profile is tested under another, appears to be less commonly reported for optical MFI; where generalisation is studied it is typically across OSNR within a fixed impairment model.

III. SYSTEM MODEL AND DATASET

A. Transmitter

We consider single-polarisation coherent transmission at $R_s = 32$ GBaud. Symbols are drawn uniformly from one of 5 constellations: QPSK, star 8-QAM, square 16-QAM, cross 32-QAM and square 64-QAM. Two of these are deliberately non-square, so that a classifier cannot succeed merely by distinguishing square grids of different densities. Every constellation is normalised to unit average power, $\mathbb{E}[|s|^2] = 1$. This normalisation is essential: without it the formats would differ in received power and any classifier could identify them trivially, whereas a real receiver applies automatic gain control and always delivers a fixed power.

B. Channel and receiver impairments

The received sample sequence is generated as

$$r[k] = \mathcal{G}\{\mathcal{I}[\mathcal{P}(\mathcal{D}\{s[k]\})]\} + n[k], \quad (1)$$

where $\mathcal{D}$ applies residual chromatic dispersion, $\mathcal{P}$ applies laser phase noise and frequency offset followed by carrier phase recovery, $\mathcal{I}$ applies receiver I/Q imbalance, $\mathcal{G}$ is automatic gain control, and $n[k]$ is circularly symmetric AWGN modelling amplified spontaneous emission.

Residual chromatic dispersion is modelled as an all-pass filter with quadratic phase, $H(\omega) = \exp\left(j\frac{\beta_2 L}{2}\omega^2\right)$ with $\beta_2 L = -\frac{\lambda^2}{2\pi c}D_{\text{tot}}$, retaining the inter-symbol interference that survives electronic compensation.

Laser phase noise is a Wiener process with per-symbol variance $2\pi\Delta\nu T_s$. A frequency offset Δf contributes an additional linear phase ramp. Both are then passed through a block-average carrier phase estimator of length 64 symbols, which removes the slowly varying component of each. What survives is the phase variation *within* one estimation block. This ordering matters: applying a frequency offset after phase recovery, as is sometimes done, leaves it uncorrected and smears each constellation point into an arc, which does not represent the output of a functioning receiver.

I/Q imbalance applies a gain mismatch between the in-phase and quadrature arms together with a departure from the ideal 90° hybrid phase, which skews the constellation rather than blurring it.

Three impairment profiles are defined — IDEAL (AWGN only), REALISTIC, and SEVERE — and are used in Section V-F. Unless stated otherwise, results use REALISTIC.

C. OSNR to SNR conversion

OSNR is an optical quantity measured in a fixed 0.1 nm (12.5 GHz) reference bandwidth across both polarisations, whereas the electrical SNR seen by the receiver follows

$$\text{SNR} = \text{OSNR} \cdot \frac{2B_{\text{ref}}}{pR_s}, \quad (2)$$

which for our parameters gives $\text{SNR}_{\text{dB}} = \text{OSNR}_{\text{dB}} - 1.07$ dB. Reporting against OSNR without this conversion overstates the electrical SNR by roughly 1 dB.

D. Simulator validation

Before any learning, the simulator is validated against theory. For a noiseless unit-power constellation the normalised fourth-order cumulant $|C_{40}|/C_{21}^2$ is an exact finite sum over the constellation points. The simulator reproduces these values for all 5 formats to within 0.004, and the three square formats additionally match the standard published constants (1.0000, 0.6800, 0.6191). We compute the reference values analytically rather than quoting them from the literature because 8-QAM and 32-QAM each have several standard variants with different cumulants.

E. Dataset

For each of the 5 formats and each of the 21 OSNR values from 5 to 25 dB in 1 dB steps we generate independent captures of 4096 symbols, giving 63,000 examples in total. Each capture is converted into three representations:

- **Raw I/Q:** the first 512 complex samples as a 2×512 real array. No human pre-processing.
- **Constellation image:** a 64×64 two-dimensional histogram of the samples over $I, Q \in [-2.5, 2.5]$, with logarithmic intensity scaling. The same spatial range is used for every format, so the zoom level carries no class information. Logarithmic scaling preserves the sparse outer constellation points, which a linear scale would quantise to zero.
- **Statistical features:** 15 numbers comprising seven normalised higher-order cumulants ($|C_{20}|$, $|C_{40}|$, $|C_{41}|$, $|C_{42}|$, $|C_{60}|$, $|C_{61}|$, $|C_{63}|$) and eight amplitude- and phase-distribution statistics.

Crucially, all three representations are derived from the *same* realisation, so any performance difference is attributable to the representation and not to differing data.

Data are split 70/15/15 into training, validation and test sets, stratified jointly on (format, OSNR) so that each of the 5×21 cells is divided in the same proportion. The test set is used exactly once.

IV. CLASSIFIERS AND REPRESENTATIONS

We evaluate five classifiers, chosen so that the model and the representation can be varied independently (Table I).

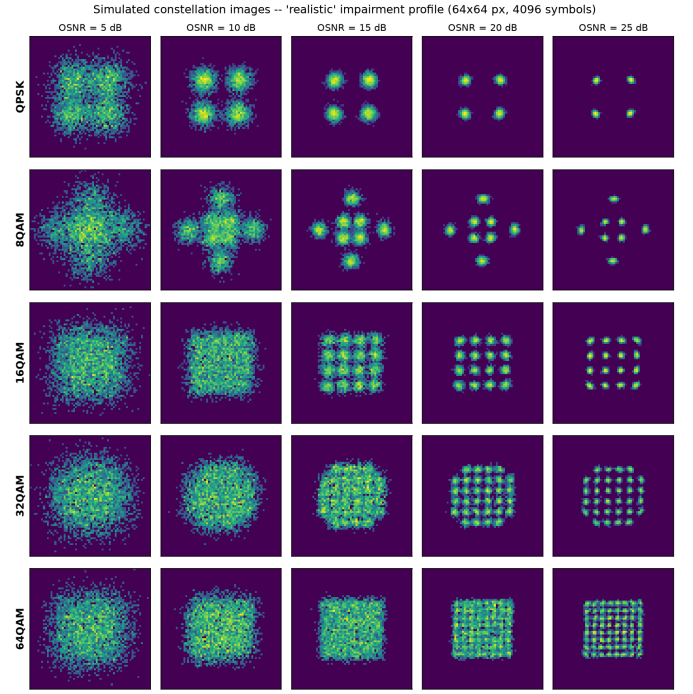


Fig. 1. Example constellation images under the REALISTIC impairment profile. Denser constellations require substantially higher OSNR to remain resolvable, which is the physical mechanism behind the accuracy–OSNR curves of Fig. 3.

A. Image representation: 2-D CNN

Three convolutional blocks (Conv3 $\times$ 3–BN–ReLU–MaxPool2) with 16, 32 and 64 channels, then dropout and two fully connected layers. After three pooling stages one feature-map element covers roughly a 30×30 region of the input, which matches the scale of the constellation structure to be recognised. Features are flattened rather than globally average-pooled, because the geometric *arrangement* of clusters in the I/Q plane is precisely the discriminative information; discarding position would discard the signal.

B. Raw I/Q representation: 1-D CNN

Three 1-D convolutional blocks (kernels 7, 5, 3; 32, 64, 64 channels) followed by global average pooling. Here pooling over position is appropriate, because the symbol stream is stationary and absolute position within the capture carries no information. This model represents the zero-pre-processing extreme.

C. Feature representation: MLP, SVM, k -NN

The MLP has hidden layers of 128 and 64 units with batch normalisation and dropout. The SVM uses an RBF kernel with $C = 10$; the k -NN uses $k = 5$. All three are preceded by standardisation whose statistics are estimated on the training partition only, to avoid leakage. The MLP is the critical control: it is a trained deep network that sees only the hand-crafted features, so comparing it against the CNN isolates

TABLE I
OVERALL TEST ACCURACY BY INPUT REPRESENTATION (MEAN $\pm$ S.D.
OVER 3 SEEDS).

Model	Input representation	Accuracy (%)	Params
CNN	64×64 image	98.73 ± 0.12	285,941
IQ-CNN	512 raw I/Q samples	86.56 ± 4.64	27,781
MLP	15 features	93.08 ± 0.10	11,013
SVM	15 features	93.41 ± 0.10	—
k -NN	15 features	91.58 ± 0.08	—

the representation, while comparing it against the SVM and k -NN isolates the algorithm.

Networks are trained with Adam ($\eta = 10^{-3}$, weight decay 10^{-4}), batch size 128, learning-rate halving on plateau, and early stopping on validation accuracy. All experiments are repeated over 3 random seeds and reported as mean $\pm$ standard deviation.

V. RESULTS

A. Overall comparison

Table I reports overall test accuracy. The image-based CNN attains $98.73 \pm 0.12\%$. The three feature-based classifiers attain 93.08%, 93.41% and 91.58% respectively, and the raw-I/Q 1-D CNN attains 86.56%.

B. Representation versus algorithm

Table I and Fig. 2 report the main comparison. Three learning algorithms of very different type (a trained deep network, a kernel method, and a non-parametric nearest-neighbour rule) operating on identical inputs span only 1.83 percentage points. Replacing the input with the constellation density image, while keeping the model a convolutional network of comparable scale, gains 5.32 points over the best of the three.

The choice of algorithm therefore accounts for 1.83 points of variation, and the choice of representation for 5.32 points, a ratio of approximately 2.9. We take this as evidence that the representation is the dominant design variable for this task.

The raw-I/Q model is both the weakest and the least stable, with a standard deviation across seeds of 4.64 points against ≤ 0.12 for the other four models. Its exact position is therefore treated as uncertain, and only its ordering (worst in all 3 runs) is relied upon below.

That the raw-I/Q representation performs worst, despite discarding no information, can be explained by the statistics of the signal. The transmitted symbols are independent and identically distributed, so temporal order carries little information about the format, whereas a 1-D convolution is structured to exploit order. The network must therefore estimate a distribution from a sequence. A 2-D histogram is itself a density estimate, so binning performs part of that estimation before training begins. We attribute the observed gain to this effect rather than to model capacity.

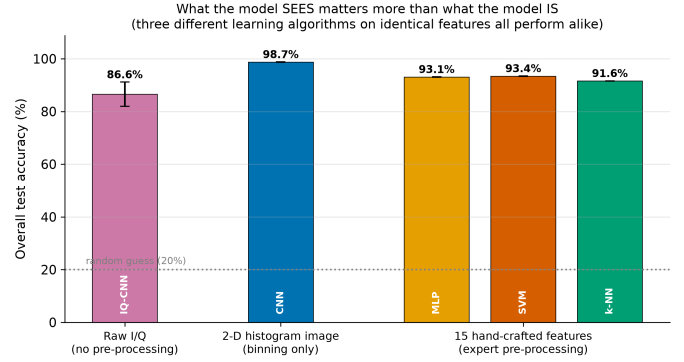


Fig. 2. Accuracy grouped by input representation. Three different learning algorithms on identical hand-crafted features perform almost identically, whereas changing the representation produces a large gain.

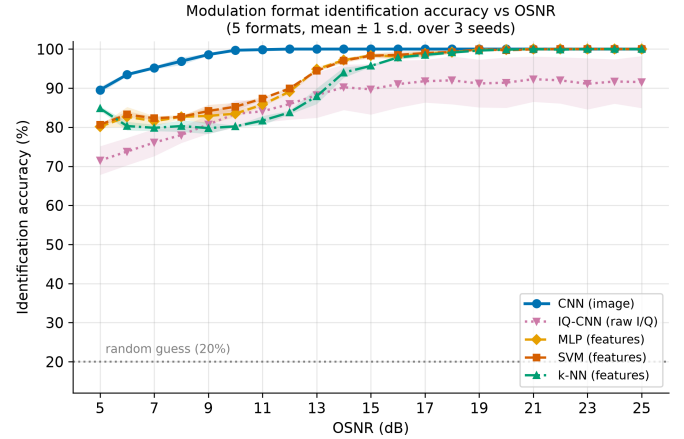


Fig. 3. Identification accuracy versus OSNR. Shaded bands are $\pm$ one standard deviation over 3 seeds. The dotted line is the 20% chance level for 5 classes.

C. Accuracy versus OSNR

Fig. 3 shows accuracy against OSNR. All curves decrease with decreasing OSNR, as they must: at low OSNR the constellation structure is physically destroyed and no classifier can recover information the channel has erased. The separation between methods is concentrated in the low-OSNR regime: averaged over 5–10 dB the CNN attains 95.57% against 83.05% for the SVM, whereas over 20–25 dB the CNN and all three feature-based classifiers exceed 99.94%, becoming effectively indistinguishable. The raw-I/Q model is the exception, plateauing at 91.68% even at high OSNR.

Expressed as an OSNR margin — the more meaningful quantity for a link budget — the CNN reaches 95% accuracy at 7 dB against 14 dB for the SVM, an advantage of 7 dB. A margin of this size translates directly into additional reach or fewer inline amplifiers.

D. Generalisation to unseen OSNR

A model trained and tested on the same discrete OSNR grid might merely have memorised the appearance of each level. To rule this out we retrain using only even-valued OSNRs and test

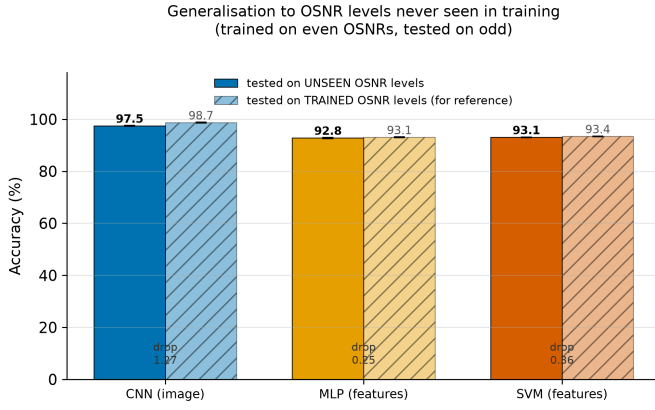


Fig. 4. Accuracy on OSNR levels withheld from training.

only on odd-valued ones, so that every test condition is unseen. The CNN attains $97.45 \pm 0.16\%$, a drop of only 1.27 points relative to the matched case (Fig. 4). The model therefore interpolates across noise levels rather than memorising them.

E. Complexity and capture length

An MFI block must eventually fit within a receiver's processing budget, so accuracy alone is not the relevant figure of merit. Fig. 5 sweeps network depth and width.

Neither has much effect. Across all six variants, accuracy spans only 1.26 percentage points (97.11–98.37%), despite the largest configuration having $9.2\times$ more parameters and $49\times$ more arithmetic than the smallest. Holding width fixed at 16 channels, depth gives 97.94%, 98.15% and 98.22% for two, three and four blocks respectively: a monotonic but negligible gain. The smallest configuration tested (67,457 parameters, 0.80 MMAC per inference) reaches 98.37%, against 98.28% for the largest (617,573 parameters).

No optimum is claimed. This sweep used a single seed, and the non-monotonic behaviour across widths indicates run-to-run variation of the same order as the differences being measured. The conclusion supported by these data is therefore the weaker one, that model capacity is not the binding constraint within this family. This is consistent with the representation result of Section V-B, in that a small network suffices once the input is a density image, and it indicates that an MFI block of well under 10^5 parameters is a realistic deployment target.

Fig. 6 reports accuracy against capture length. Accuracy is largely retained down to 512 symbols (94.08%), compared with 97.84% at 4096 symbols. Short captures are therefore sufficient, which relaxes the memory requirement of a practical monitor.

F. Robustness to unseen impairments

Finally we test transfer across channel conditions (Fig. 7), training on one impairment profile and testing on another. This is the setting that matters operationally: a model is trained on a channel model, and then deployed on a link that may be worse than assumed.

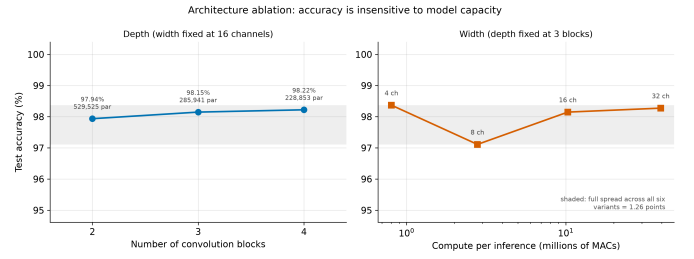


Fig. 5. Architecture ablation: accuracy against depth and against compute.

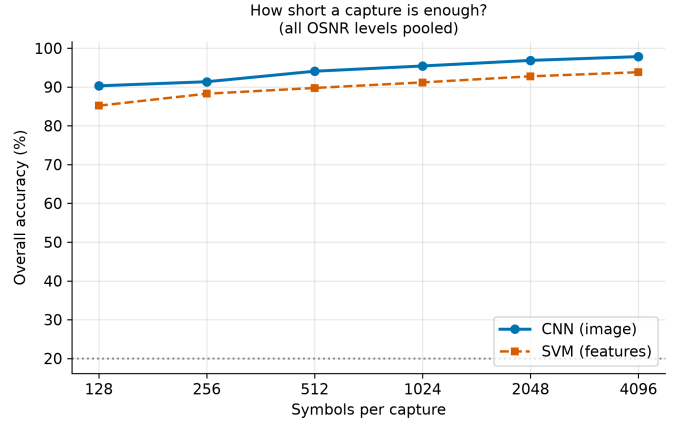


Fig. 6. Accuracy against the number of symbols in the capture.

The two representations behave very differently under mismatch. Trained on the IDEAL AWGN-only channel and tested on SEVERE, the CNN falls from 97.92% to 89.78%, a drop of 8.13 points. Under the same conditions the feature-based SVM falls from 93.02% to 21.84%, a drop of 71.18 points. With 5 classes the chance level is 20%, so the feature-based classifier retains essentially no discriminative ability.

A plausible explanation is that higher-order cumulants are statistics of an assumed signal model. I/Q imbalance, residual dispersion and phase error all shift their values, so the decision boundaries learned in the 15-dimensional feature space no longer correspond to the same regions of signal space. The CNN instead operates on the spatial structure of the density image, where a distorted constellation continues to resemble its undistorted form more closely than it resembles any other format, and degradation is correspondingly gradual.

Training on the REALISTIC profile recovers much of the loss for both models (91.17% and 71.98% respectively on SEVERE), confirming that the channel model used during training materially determines deployed performance.

This result extends the contribution of Section V-B. The advantage of the density representation is not confined to matched conditions; it is larger still under channel mismatch, which is the condition a simulation-trained monitor encounters on deployment. It also suggests that matched-channel accuracy, which is what the MFI literature generally reports, can substantially overstate the performance available in practice.

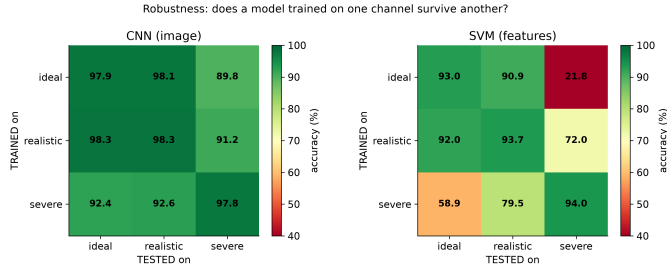


Fig. 7. Cross-channel robustness. Rows: training profile. Columns: test profile.

G. Does the model know when it is failing?

Section V-F reports accuracy under channel mismatch, but accuracy is visible only to the experimenter. A deployed monitor knows only what it predicted and how confident it was. The operationally relevant question is therefore whether the model’s own confidence signals its failure, and whether gating on that confidence keeps the monitor usable.

We take the maximum softmax probability as the confidence score, this being the quantity a deployed system would threshold on, and use Platt scaling to obtain the equivalent for the SVM. Models are trained on the IDEAL channel and evaluated on SEVERE.

Under matched conditions all three models are well calibrated, with mean confidence within one percentage point of accuracy. Under mismatch they diverge sharply (Fig. 8). The CNN falls to 89.78% accuracy while reporting 95.67% confidence, an overconfidence of 5.89 points. The SVM falls to 22.41% — barely above the 20% chance level — while reporting 42.95% confidence, an overconfidence of 20.54 points. It claims roughly twice the accuracy it achieves.

Confidence nonetheless remains informative in the ranking sense for both models: the AUROC of confidence as a detector of the model’s own errors is 0.947 for the CNN and 0.949 for the SVM. Gating on confidence should therefore restore reliability. The cost of doing so is where the representations separate.

At a 90% confidence gate on the SEVERE channel, the CNN still answers 86.9% of inputs, at 96.77% accuracy on those it answers. The SVM answers 0.2% — a coverage ratio of approximately $447\times$. The feature-based monitor can be made accurate only by silencing it: at that operating point it responds to approximately two inputs in every thousand. Under the milder REALISTIC-to-SEVERE shift the same effect appears in weaker form, with SVM coverage of 5.3% at the same gate.

We therefore state the contribution of Section V-B in a stronger form. The input representation determines not only accuracy under matched conditions, but whether a confidence-gated monitor remains *operational* under mismatch at all. To our knowledge this interaction between input representation and failure self-detectability has not previously been reported; uncertainty quantification for modulation classification has been studied for a single representation [10], and representation comparison has been studied for accuracy alone [7].

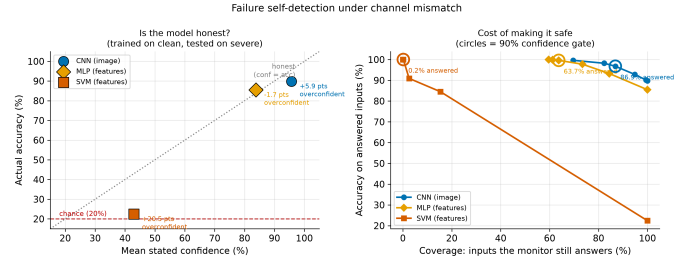


Fig. 8. Failure self-detection under channel mismatch. Left: stated confidence against actual accuracy on the unseen SEVERE channel; the dotted diagonal marks an honest model. Right: the accuracy–coverage trade-off under confidence gating, with circles marking the 90% gate.

H. Impairment tolerance budget

Sections V-F and V-G compare discrete channel profiles, which establishes that mismatch is harmful but not how much mismatch is affordable. We therefore vary one impairment at a time, holding the remaining four at their training values, and record accuracy as that parameter moves away from the value the model was trained on (Fig. 9). Models are trained once on the REALISTIC channel; no retraining is performed.

Table II reports the largest tested deviation at which each model still achieves 90% accuracy. Three observations follow.

First, the two representations do not fail through the same mechanism. The feature-based classifier is highly sensitive to the two impairments that distort constellation *geometry*: accuracy falls from 93.11% to 20.35% as I/Q gain imbalance increases from 0.3 dB to 2.5 dB, and from 93.11% to 20.10% as I/Q phase skew increases from 2° to 14° — the 20% chance level in both cases. The CNN retains 95.48% and 96.35% respectively at the same extremes. This is consistent with the mechanism proposed in Section V-F: cumulants are moments computed under an implicit assumption of balanced quadratures, so a geometric skew corrupts them directly, whereas a skewed constellation image remains recognisable as a skewed version of the correct constellation.

Second, and contrary to the pattern elsewhere in this paper, the feature-based classifier is *more* tolerant of residual chromatic dispersion: at 150 ps/nm it retains 80.14% against 54.89% for the CNN. This is also consistent with the same mechanism. Dispersion-induced inter-symbol interference broadens the symbol cloud in a manner closer to added noise than to geometric distortion, and higher-order cumulants are specifically constructed so that Gaussian contributions cancel. We report this because it qualifies the paper’s central claim: the density representation is not uniformly superior, and its advantage is specific to distortions that preserve gross constellation structure while altering its geometry.

Third, laser linewidth is essentially irrelevant to both models over the entire range tested, up to 2 MHz, because block-average carrier phase recovery removes it. Residual frequency offset affects both models identically, with a sharp common failure between 50 MHz and 100 MHz.

The practical reading is that a designer intending to train

TABLE II
IMPAIRMENT TOLERANCE BUDGET: LARGEST TESTED DEVIATION STILL MEETING 90% ACCURACY. ARROWS MARK THE MORE TOLERANT REPRESENTATION.

Impairment	CNN	SVM	
Laser linewidth	>2000 kHz	>2000 kHz	equal
Frequency offset	50 MHz	50 MHz	equal
I/Q gain imbalance	>2.5 dB	0.6 dB	CNN $\geq 4\times$
I/Q phase skew	>14°	4°	CNN $\geq 3.5\times$
Residual dispersion	50 ps/nm	90 ps/nm	SVM $1.8\times$

an MFI monitor on a simulated channel must characterise I/Q imbalance far more accurately than laser linewidth, and that the required accuracy depends on the representation: approximately $4\times$ tighter for a cumulant-based monitor than for an image-based one.

VI. DISCUSSION

The results suggest a practical design rule for MFI: invest in the input representation before investing in model capacity. A modest CNN on a density image outperforms both a deeper-in-spirit model on raw samples and three different learners on expert features, and the ablation shows that most of its capacity is not needed.

A likely mechanism is the following. The discriminative information in MFI is the distribution of received symbols in the complex plane. A two-dimensional histogram is a direct non-parametric estimate of that distribution, so the CNN begins from a near-sufficient statistic and need only classify its shape. Cumulants are also statistics of the same distribution, but constitute a lossy 15-dimensional projection of it chosen in advance. The raw sample stream retains all of the information, but presents it in a time-indexed form from which the density must be re-estimated by the network.

Limitations. The evaluation is simulation-based; no experimental captures are used. We model single-polarisation transmission, a single symbol rate, and a linear channel — fibre nonlinearity, polarisation-mode dispersion and polarisation multiplexing are not included. The RBF-SVM is trained on a capped subsample for tractability. Extending the study to experimental data, to polarisation-multiplexed transmission and to joint format-and-OSNR estimation are natural next steps.

VII. CONCLUSION

We separated the contributions of model and representation in machine-learning modulation format identification. On a common benchmark of 63,000 captures spanning 5 formats and a realistically impaired channel, varying the learning algorithm over identical inputs changed accuracy by 1.83 percentage points, while varying the input representation changed it by 5.32 points, a ratio of approximately 2.9. A trained multilayer perceptron on hand-crafted features did not outperform a kernel machine on the same features, which indicates that model capacity was not the limiting factor.

A compact CNN on constellation density images achieved 98.73% overall, an OSNR margin of 7 dB over the best feature-based classifier, and 97.45% on OSNR values withheld from training. Its largest advantage occurred under channel mismatch, where the feature-based classifier fell to the 20% chance level while the CNN retained 89.78%.

We conclude that the input representation, rather than model capacity, is the dominant design variable for this task; that the benefit is one of robustness as much as accuracy; and that the channel model used during training substantially determines whether either transfers.

REPRODUCIBILITY

All source code, the data-generation simulator, trained model weights and the scripts that produce every figure and number in this paper are available at <https://github.com/techi101/optical-mfi-representation>. All experiments run on a single CPU; no GPU is required. Random seeds are fixed and an automated verification script re-derives the reported accuracies from the saved model weights.

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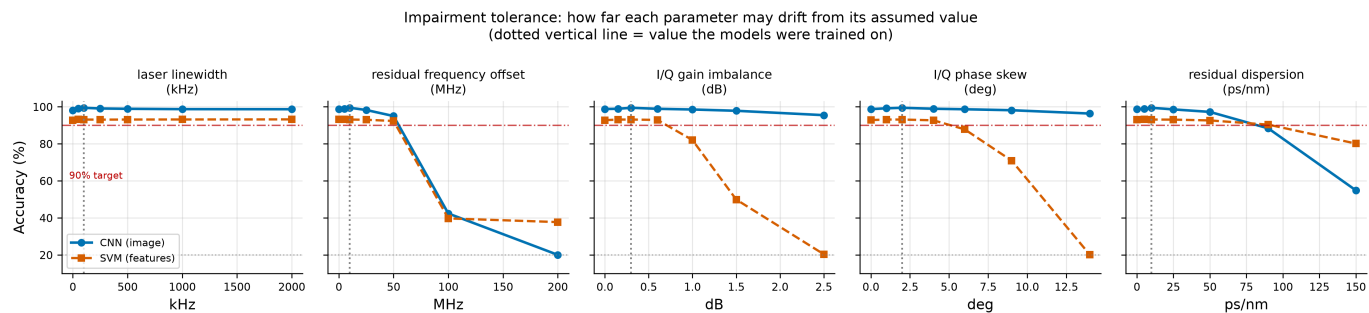


Fig. 9. Impairment tolerance. Each panel sweeps one impairment while the others are held at their training values. The dotted vertical line marks the trained value; the dash-dotted horizontal line marks a 90% accuracy target.